THE PROBLEM

When the towers go down, rescue goes blind

Every cyclone and flood season, coastal and low-lying districts lose mobile coverage. Towers lose power, backhaul links are cut and the remaining network is overloaded. This happened in the 2015 Chennai floods and again during Cyclone Michaung in 2023.

People cannot call for help

Trapped families, the elderly, patients and people with disabilities cannot call, text or share their location.

Control rooms work blind

There is no view of who needs help, where they are, how many people are with them, or which case is most urgent.

Rescue capacity is misdirected

Boats and teams search by sight and word of mouth, so they reach the easiest-to-see rather than the most critical.

Problem statement

When cyclones and floods take down mobile towers and power, people in danger cannot call for help, and district control rooms cannot see who needs rescue, where they are, or how urgent it is.

PROPOSED SOLUTION

A complete rescue loop on phones people already own

MeshAlive is an Android app and a browser-based district dashboard. Together they form a two-way emergency channel that works without towers, internet or extra hardware.

1

Ask for help

One-tap SOS with GPS and a situation report, scored 0–100

2

Phones relay

Bluetooth and Wi-Fi mesh, battery-aware, most urgent first

3

Rescuers carry

Boats and teams carry SOS out of areas with no route

4

Control room sees

First phone with signal uploads to a live dashboard

5

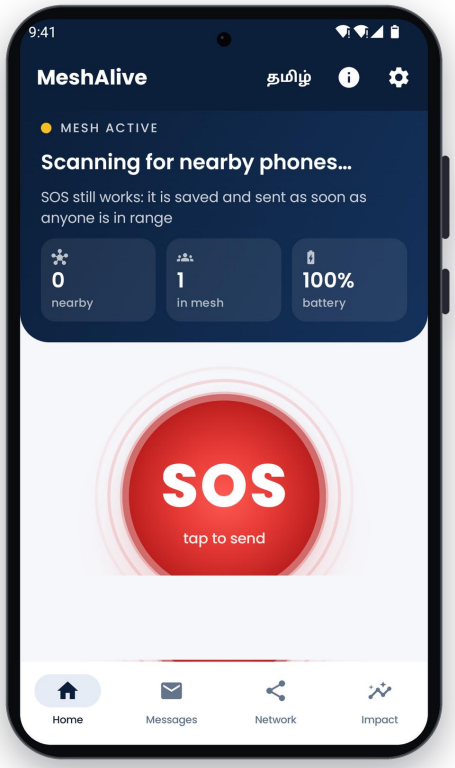
Help returns

Dispatch and evacuation alerts travel back, signed

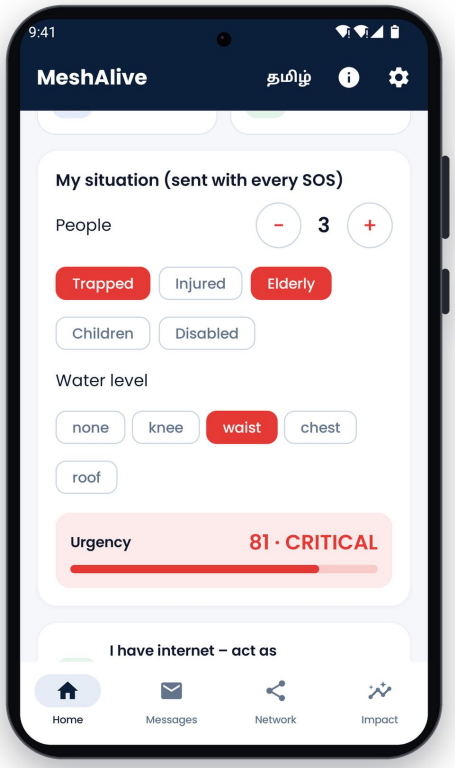
- No towers, no internet and no SIM needed for the mesh
- No extra hardware: uses citizens’ and rescuers’ existing phones
- Works in Tamil and English, with large, simple controls

PROTOTYPE

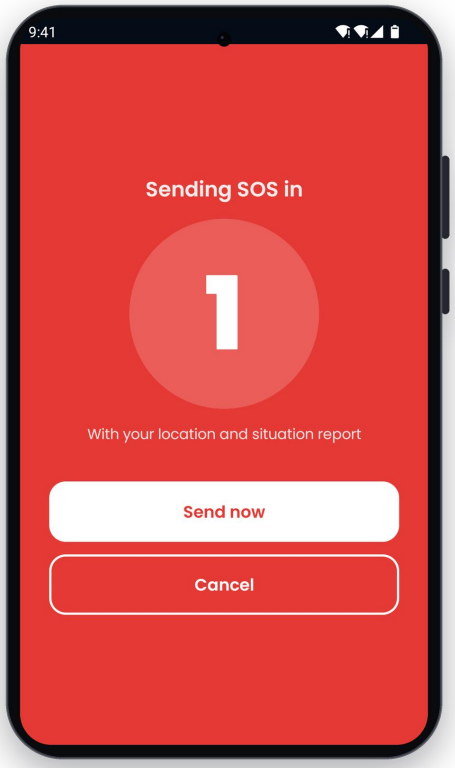
Mobile app: working Android prototype



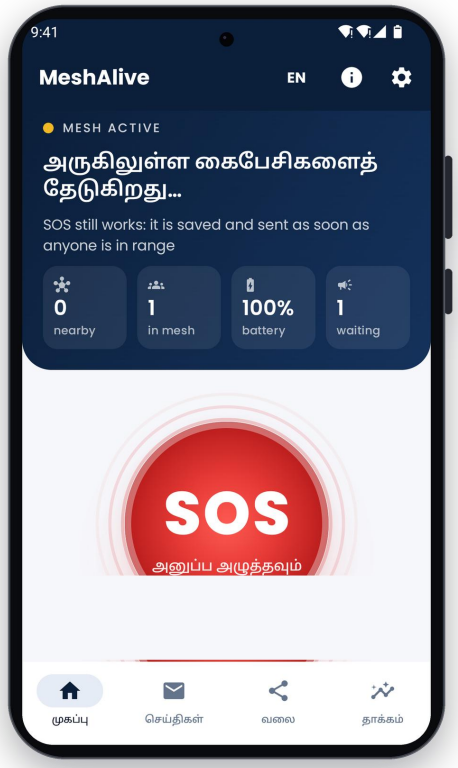
Home: mesh status, one-tap SOS



Situation report: urgency 81



3-second countdown



Complete flow in Tamil

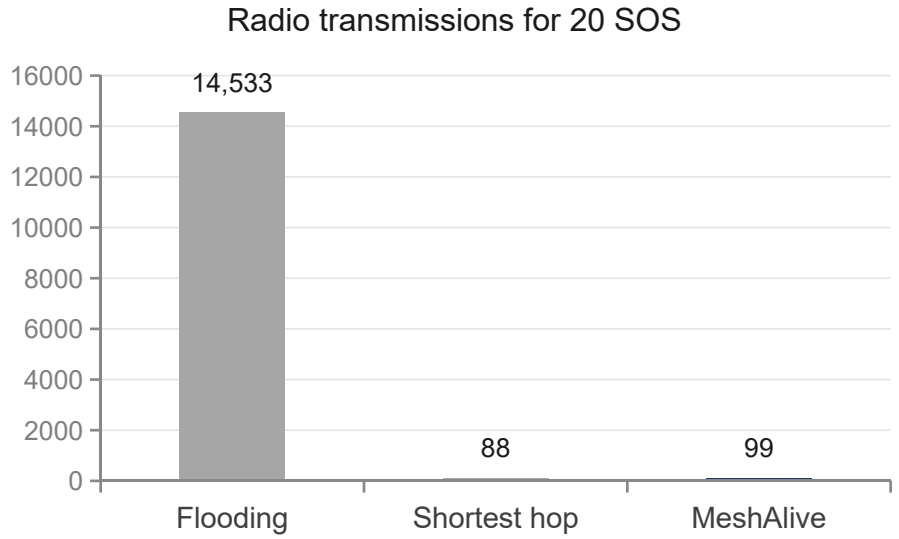
Built with Kotlin and Jetpack Compose. All screens are captured from the running app.

EVIDENCE

Results: 200-phone simulation

Measured with the app's own routing engine: 200 phones, 20 SOS messages, the same network for every method.

Measure	Flooding	Shortest hop	MeshAlive
SOS delivered	85%	100%	**100%**
Radio transmissions	14,533	88	**99**
Average hops	5.47	4.40	**4.95**
Battery left, weakest phone	0.0%	3.2%	**6.3%**



- Every SOS delivered, with 147 times fewer transmissions than flooding
- Twice the battery left on the weakest phone compared with shortest-hop routing, so the mesh stays alive longer
- 49 automated tests cover routing, forwarding, triage, command signatures, the planner and evidence reports

INNOVATION

What is new

Capability	Chat apps	LoRa radios	MeshAlive
Reaches the district control room	No	No	**Yes, live dashboard**
Reply back to the person	No	No	**Dispatch and evacuation alerts**
Prioritises urgent cases	No	No	**Urgency score 0–100**
No phone in range	Message waits	Message waits	**Rescue carriers carry it out**
Battery management	Flooding	Flooding	**Battery-aware routing**
Protection against fake alerts	Not addressed	Not addressed	**Signed commands**
Extra hardware	None	₹2,500+ per person	**None**
Indian languages	English	English	**Tamil and English**

IMPLEMENTATION

Implementation roadmap

Phase 1 · Field pilot 3 months

- One flood-prone ward with the DDMA
- NDRF/SDRF boats as rescue carriers
- Dashboard in the district EOC
- Measure delivery rate and SOS-to-dispatch time

Phase 2 · District rollout Pre-monsoon

- Install drives via schools, colleges, panchayats and SHGs
- Planner sets the install target per ward
- Training for control-room operators

Phase 3 · State and national Scale-up

- Distribution through the SDMA
- Dashboard feeds into state EOCs
- Alignment with the Common Alerting Protocol
- More Indian languages; iOS version

Success measures: SOS delivery rate, time from SOS to dispatch, households confirmed safe, and installs per ward.